

VED GUND, PHD

vgund@wesleyan.edu — 265 Church St, Room 543, Middletown, CT - 06459 — Google Scholar

EDUCATION

Cornell University Ph.D., Electrical and Computer Engineering <i>Thesis: Hybrid MEMS for Transient Microsystems and Chip-Scale Gas Sensors</i> <i>Committee: Amit Lal (chair), Debdeep Jena, Chris Ober, Al Molnar</i>	<i>2011-2017</i>
Stanford University M.S., Electrical Engineering	<i>2010-2011</i>
Stanford University B.S., Electrical Engineering	<i>2007-2011</i>

EMPLOYMENT

Assistant Professor <i>College of Design and Engineering Studies, Wesleyan University</i>	<i>2026-present Middletown, CT</i>
MEMS Research Scientist - Advanced Technology <i>Honeywell Aerospace</i>	<i>2023-2026 Plymouth, MN</i>
Research Associate - SonicMEMS Lab <i>School of Electrical and Computer Engineering, Cornell University</i>	<i>2020-2023 Ithaca, NY</i>
Senior Process Engineer - Technology Research <i>Intel Corporation</i>	<i>2017-2020 Hillsboro, OR</i>
Graduate Research Assistant - SonicMEMS Lab <i>School of Electrical and Computer Engineering, Cornell University</i>	<i>2011-2017 Ithaca, NY</i>
Summer REU and Undergraduate Research Assistant - VLF Lab <i>Department of Electrical Engineering, Stanford University</i>	<i>2008-2011 Stanford, CA</i>

HONORS AND AWARDS

• Outstanding Engineering Award , Honeywell Aerospace	<i>2025</i>
• Top Flyer Award , Honeywell Aerospace	<i>2024</i>
• Divisional Recognition Award , Intel Corporation	<i>2019</i>
• Sigma Xi, The Scientific Research Honors Society	<i>2017</i>
• Colman Inclusive Leadership Program Fellow	<i>2017</i>
• Corning Inc. Best Paper Award , CNF Annual Meeting	<i>2015</i>
• Cornell Fellowship , Cornell University	<i>2011-2012</i>
• Analog Devices Design Project Finalist , Stanford University	<i>2010</i>
• Tau Beta Pi Engineering Honors Society , California Gamma Chapter	<i>2010</i>
• Stanford Fund Scholarship for Undergraduates , Stanford University	<i>2007-2011</i>

TEACHING EXPERIENCE

Course Instructor ECE4320 - Integrated Micro Sensors and Actuators: Bridging the Physical and Digital Worlds	2020-2023
<i>School of Electrical and Computer Engineering, Cornell University</i>	<i>Ithaca, NY</i>
Lab Instructor ECE5960 - Piezoelectric and Ultrasonic Waves Microsystems for Sensing, Actuation, and Physical Computing	2020-2022
<i>School of Electrical and Computer Engineering, Cornell University</i>	<i>Ithaca, NY</i>
Teaching Assistant ECE4320 - MicroElectro Mechanical Systems (MEMS)	2016-2017
<i>School of Electrical and Computer Engineering, Cornell University</i>	<i>Ithaca, NY</i>

MENTORSHIP

Shubham Jadhav, Cornell University <i>PhD Student - Ferroelectric NEMS switches</i>	2020-2023
Landon Ivy, Cornell University <i>PhD Student - Flexible printed electronics for AI</i> <i>Current Affiliation: Staff Researcher, Sandia National Labs, NM</i>	2020-2023
Di Ni, Cornell University <i>PhD Student - Electrostatic actuators using pyroelectric lithium niobate</i> <i>Current Affiliation: Assistant Professor, Lehigh University</i>	2020-2022
Ujjwal Singh, Cornell University <i>M.Eng Student - 3D-printed pyroelectric high voltage generation</i> <i>Current Affiliation: Physical Design Engineer, AMD</i>	2021-2022
Sonal Lodha, Cornell University <i>M.Eng Researcher - High voltage generation with pyroelectric materials</i> <i>Current Affiliation - Technical Services, Epic Systems</i>	2016-2017
Nathan Lambert, Cornell University <i>Undergraduate Researcher - Lithium niobate ionizer for chip-scale gas sensing</i> <i>Current Affiliation - Research Scientist, Allen Institute for AI</i>	2016-2017
Zachary Schaffer, SUNY Polytechnic Institute <i>NSF Research Experience for Undergraduates - Piezoresistive graphene accelerometer</i> <i>Current Affiliation - NPI Engineer, Broadcom</i>	2015
Manikandan Srinivasan, IIT-Madras <i>Summer Undergraduate Research Intern - MEMS piezoelectric transformers</i> <i>Current Affiliation - Quantitative Strategist, Goldman Sachs</i>	2014

GRANTS & FUNDING

Honeywell Aerospace

MEMS for hardware security 2023-2024
co-PI Internal Kickstarter Funding - \$300,000

Design of MEMS ferroelectric and PCM computing engines 2023
PI Internal Seed Funding - \$5,000

Cornell University

Long Term Reliable Neural Recordings & Neuro Modulation Using GHz Ultrasonics 2020-2021
Technical Design - \$375,000

SEMI FlexTech: AI for Printed Electronics Assisted by GHz Ultrasonic Metrology 2020-2023
Grant Writing, Technical Design - \$524,504

SoftMEMS: Student Research Project - Cornell ECE4320 2020-2021
Course Instructor, Project Design - Software award \$8,000

Intel ISRA: Broad-band chemical sensors for ubiquitous cell phone applications 2014-2016
Grant Preparation, Student Technical Lead - \$200,000

DARPA: Vanishing Programmable Resources (VAPR) Program 2013-2017
Grant Preparation, Student Technical Lead - \$1,900,000

SERVICE

- **Reviewer** - Advanced Functional Materials, Advanced Electronic Materials, IEEE Journal of Electron Devices Society, IEEE Transactions on Electron Devices, IEEE Electron Device Letters, IEEE Sensors Conference, IEEE Sensors Journal, IEEE Journal of Microelectromechanical Systems, IEEE Biosensors Conference, Journal of Instrumentation
- **Volunteer** - USA Astronomy and Astrophysics Organization 2023-present
- **Intel STEM Volunteer** - National Engineer's Week 2019
- **Judge** - Oregon State Science Fair Project 2018
- **Cleannroom Tour Guide** - STEM Outreach Cornell Nanoscale Facility 2015-17
- **Judge** - Cornell University ECE Day 2014-16
- **President** - Cornell Electron Devices Society (EDS) 2014
- **Meet Volunteer** - Fingerlakes Runners Club Track Meets 2015-17

JOURNAL PUBLICATIONS

1. S. Sandeep, R.M.R. Pinto, J. Rudresh, **V. Gund**, K.K. Nagaraja, K.B. Vinayakumar, "Piezoelectric aluminum nitride thin films for CMOS compatible MEMS: Sputter deposition and doping", *Critical Reviews in Solid State and Materials Sciences*, 1-28, 2025
2. K. Nomoto, J. Casamento, T.-S. Nguyen, L. Li, H. Lee, C. Savant, A. Hickman, T. Maeda, J. Encomendero, **V. Gund**, T. Vasen, S. Afroz, D. Hannan, J. Hwang, D. Jena, H. G. Xing, "AlScN/GaN HEMTs with 4 A/mm on-current and maximum oscillation frequency > 130 GHz", *Applied Physics Express*, 2025
3. C. Savant, **V. Gund**, K. Nomoto, T. Maeda, S. Jadhav, M. Ramesh, E. Kim, T.-S. Nguyen, Y.-H. Chen, J. Casamento, A. Lal, H. Xing, D. Jena, "Ferroelectric AlBN Films by Molecular Beam Epitaxy", *Applied Physics Letters*, 2024 (**Editor's Pick and Cover Article for August 2024**)
4. S. Jadhav, **V. Gund**, B. Davaji, D. Jena, H. G. Xing, and A. Lal, "Graphene HZO-based FerroNEMS MAC for In-Memory Computing", *Applied Physics Letters*, 120.15, 2022.
5. R.M.R. Pinto, **V. Gund**, R. A. Dias, K. K. Nagaraja, K. B. Vinayakumar, "CMOS-Integrated Aluminum Nitride for MEMS Applications: A Review", *Journal of Microelectromechanical Systems*, 2022.
6. J. Casamento, H. Lee, T. Maeda, **V. Gund**, L. van Deurzen, A. Lal, D. Jena, and H. G. Xing, "Epitaxial $Sc_xAl_{1-x}N$ on GaN exhibits attractive high-K Dielectric Properties", *Applied Physics Letters*, no. 15, 152901, 2022.
7. R. M. R. Pinto, **V. Gund**, C. Calaza, K. K. Nagaraja, and K. B. Vinayakumar, "Piezoelectric Aluminum Nitride Thin-Films: A Review of Wet and Dry Etching Techniques", *Microelectronic Engineering*, 111753, 2022.
8. J. Casamento, **V. Gund**, H. Lee, K. Nomoto, T. Maeda, B. Davaji, M.J. Asadi, J. Wright, Y.-T. Shao, D. A. Muller, A. Lal, H. Xing, and D. Jena, "Ferroelectricity in Polar ScAlN/GaN Epitaxial Semiconductor Heterostructures", *arXiv:2105.10114*, 2021.
9. K. B. Vinayakumar* **V. Gund***, and A. Lal, "Miniaturized Low Power Vacuum Pump Using Pyroelectric Lithium Niobate", *Journal of Vacuum Science and Technology B, Nanotechnology and Microelectronics: Materials, Processing, Measurement, and Phenomena*, 39 (2), 024202, 2021.
10. **V. Gund**, A. Ruyack, A. Leonardi, V. Kumar, C. K. Ober, and A. Lal, "Spatially Controlled Transience of Graphene-Polymer Electronics with Silicon Singulation", *Advanced Functional Materials*, 29 (20), 1900592, 2019.
11. **V. Gund**, A. Ruyack, S. Ardanuc, and A. Lal, "Graphene Stress Transducer-Based Thermo-Mechanically Fractured Micro Valve", *Journal of Microelectromechanical Systems*, 27 (3), 555-569, 2018.

REFERRED CONFERENCE PROCEEDINGS

1. **(invited)** J. Reinke, M. Gnerlich, A. Brown, **V. Gund**, D. Effa, "MEMS Gyroscope Errors During Vibration", *2026 IEEE International Symposium on Inertial Sensors and Systems (INERTIAL)*. IEEE, 2026
2. **(invited)** J. Casamento, K. Nomoto, T.-S. Nguyen, H. Lee, C. Savant, L. Li, A. Hickman, T. Maeda, Y.-T. Shao, J. Encomendero, **V. Gund**, T. Vasen, S. Afroz, D. Hannan, D. Muller, H. G. Xing, and D. Jena, "AlScN High Electron Mobility Transistors: Integrating High Piezoelectric, High K Dielectric, and Ferroelectric Functionality", *IEEE BiCMOS and Compound Semiconductor Integrated Circuits and Technology Symposium (BCICTS)*, 2023.

3. S. Jadhav, **V. Gund**, M. Ramesh, D. Jena, and A. Lal, "Design and Implementation of an AlScN-Based FeMEMS Multiplier for In-Memory Computing Applications", *IEEE International Symposium on Applications of Ferroelectrics (ISAF)*, 2023.
4. L. Ivy, Y. Xie, T. Lobo, **V. Gund**, B. Davaji, M. Garud, P. Doerschuk, and A. Lal, "Feature-Based Machine Learning for Predicting Resistances in Printed Electronics", *IEEE International Conference on Flexible, Printable Sensors and Systems (FLEPS)*, 2023.
5. S. Jadhav, **V. Gund**, and A. Lal, "Programmable Ferroelectric HZO NEMS Mechanical Multiplier for in-Memory Computing.", *IEEE International Conference on Micro Electro Mechanical Systems (MEMS '23)*, 2023.
6. J. Casamento, K. Nomoto, T.S. Nguyen, H. Lee, C. Savant, L. Li, A. Hickman, T. Maeda, J. Encomendero, **V. Gund**, A. Lal, J. C. M. Hwang, H. G. Xing, and D. Jena. "FerroHEMTs: High-Current and High-Speed All-Epitaxial AlScN/GaN Ferroelectric Transistors", *IEEE International Electron Devices Meeting (IEDM)*, 2022.
7. L. Ivy, **V. Gund**, B. Davaji, C. Ospina, D. Ni, P. Doerschuk, and A. Lal, "Laser-Induced Graphene Pressure Sensors Manufactured via Inkjet PCB Printer", *IEEE International Conference on Flexible, Printable Sensors and Systems (FLEPS)*, 2022.
8. **V. Gund**, K. Nomoto, H. G. Xing, D. Jena, and A. Lal, "Intrinsically Switchable GHz Ferroelectric ScAlN SAW Resonators", *IEEE International Symposium on Applications of Ferroelectrics (ISAF)*, 2022.
9. **V. Gund**, B. Davaji, S. Jadhav, H. Lee, D. Jena, H. G. Xing, D. Jena, and A. Lal, "Multi-level Analog Programmable Graphene Resistive Memory with Fractional Channel Ferroelectric Switching in Hafnium Zirconium Oxide", *IEEE Frequency Control Symposium (FCS)*, 2022.
10. D. Ni, **V. Gund**, and A. Lal, "Kilo-Volt Pyroelectric Voltage Generation and Electrostatic Actuation with Fluidic Heating", *Hilton Head Solid-State Sensors, Actuators and Microsystems Workshop*, 2022.
11. L. Ivy, **V. Gund**, B. Davaji, C. Ospina, D. Ni, P. Doerschuk, and A. Lal, "Laser-Induced Graphene Pressure Sensors Manufactured via Inkjet PCB Printer", *Hilton Head Solid-State Sensors, Actuators and Microsystems Workshop*, 2022.
12. A. Baskota, L. Ivy, C. Ospina, J. Kuo, J.H. Hwang, **V. Gund**, B. Davaji, P. Doerschuk, and A. Lal, "In-Situ Ultrasonic Imaging of Printed Electronics Ink Deposition and Curing", *Hilton Head Solid-State Sensors, Actuators and Microsystems Workshop*, 2022.
13. R. Sanghvi, D. Ni, **V. Gund**, Q. Ji, T. Schenkel, K. K. Afridi, A. Lal, and E. Petro, "Application of Microelectromechanical-system based RF Linear Accelerators for Ultra-High Specific Impulse Ion Propulsion", *AIAA Propulsion and Energy Forum and Exposition*, 2021.
14. **V. Gund**, B. Davaji, H. Lee, M.J. Asadi, J. Casamento, H. G. Xing, D. Jena, and A. Lal, "Towards Realizing the Low-Coercive Field Operation of Sputtered Ferroelectric $Sc_xAl_{1-x}N$ ", *IEEE Conference on Solid-State Sensors, Actuators, and Microsystems (Transducers '21)*, 2021.
15. D. Ni, **V. Gund**, and A. Lal, "PCB Integrated $LiNbO_3$ Pyroelectric High-Voltage Supply with Electrostatic Switch Regulator", *IEEE Conference on Solid-State Sensors, Actuators, and Microsystems (Transducers '21)*, 2021.
16. J. Casamento, **V. Gund**, K. Nomoto, T. Maeda, B. Davaji, M. J. Asadi, J. Wright, Y. Shao, D. A. Muller, A. Lal, H. Xing, and D. Jena, "Epitaxial $Sc_xAl_{1-x}N$: Structural, Chemical, Electrical Properties, and Ferroelectric Behavior", *21st International Conference on Molecular Beam Epitaxy*, 2021.

17. **V. Gund**, B. Davaji, H. Lee, M.J. Asadi, J. Casamento, H. G. Xing, D. Jena, and A. Lal, "Temperature-dependent Lowering of Coercive Field in 300 nm Sputtered Ferroelectric $Al_{0.70}Sc_{0.30}N$ ", *IEEE International Symposium on Applications of Ferroelectrics (ISAF)*, 2021.
18. **V. Gund**, and A. Lal, "Graphene-on-Polymer Flexible Vaporizable Sensor", *IEEE International Conference on Micro Electro Mechanical Systems (MEMS '21)*, 2021.
19. Y. Liu, J. Kuo, B. Davaji, **V. Gund**, and A. Lal, "Characterization of AlScN on CMOS Wafer", *IEEE Frequency Control Symposium (IFCS)*, 2020.
20. **V. Gund**, A. Ruyack, A. Leonardi, V. Kumar, C. K. Ober, and A. Lal, "Individually detachable polymer-silicon micro-parts for vaporizable electronics", *IEEE Conference on Solid-State Sensors, Actuators, and Microsystems (Transducers '17)*, 2017.
21. V. Kumar*, **V. Gund***, N. Lambert, S. Lodha, and A. Lal, "Enhanced Lithium Niobate based pyroelectric ionization for chip-scale ion mobility-based gas sensing", *IEEE Sensors*, 2016.
22. **V. Gund**, A. Ruyack, K. Camera, S. Ardanuc, C. K. Ober, and A. Lal, "Transient micropackets for silicon dioxide and polymer-based vaporizable electronics", *IEEE International Conference on Micro Electro Mechanical Systems (MEMS '16)*, 2016.
23. **V. Gund**, A. Ruyack, S. Ardanuc, and A. Lal, "Graphene one-shot microvalve: Towards vaporizable electronics", *IEEE International Conference on Micro Electro Mechanical Systems (MEMS '15)*, 2015.
24. **V. Gund**, A. Ruyack, K. Camera, S. Ardanuc, C.K. Ober, and A. Lal, "Multi-modal graphene polymer interface characterization platform for vaporizable electronics", *IEEE International Conference on Micro Electro Mechanical Systems (MEMS '15)*, 2015.
25. A. Ruyack, **V. Gund**, K. Camera, S. Ardanuc, C.K. Ober, and A. Lal, "Alkali metal based micro combustion using graphene micro-valve trigger", *IEEE International Conference on PowerMEMS*, 2015.
26. **V. Gund**, L. Sanchez, R. Polcawich, and A. Lal, "Low-voltage thin-film piezoelectric sense-actuate fan-pair for chip-scale gas-sensors", *Solid-State Sensors, Actuators and Microsystems Workshop, (Hilton Head '14)*, 2014.
27. **V. Gund**, Y. Shi, S. Ardanuc, and A. Lal, "Low-voltage (<5V) ion-mobility spectrometer array for label-free gas detection", *IEEE Conference on Solid-State Sensors, Actuators, and Microsystems (Transducers '13)*, 2013.

(* indicates authors made equal contributions)

CONFERENCE PRESENTATIONS

1. C. P. Savant, **V. Gund**, K. Nomoto, T. Maeda, S. Jadhav, T.-S. Nguyen, J. Casamento, A. Lal, H. G. Xing, and D. Jena, "Ferroelectricity in High-K BAIN Films Grown by Molecular Beam Epitaxy", *IEEE International Symposium on Applications of Ferroelectrics (ISAF)*, 2023.
2. Qing Ji, Nicholas Valverde, Arun Persaud, Yuetao Hou, Di Ni, Ariel Amsellem, **Ved Gund**, Meera Garud, Zhihao Qin, Peter A. Seidl, Khurram K. Afridi, Amit Lal, Steve Lund, Thomas Schenkel, "Development of compact multi-beam ion accelerators for plasma heating", *American Physical Society - Division of Plasma Physics*, 2022
3. S. Jadhav, **V. Gund**, B. Davaji, D. Jena, H. G. Xing, and A. Lal, "Ultra-thin HZO-based unimorph beams for Biosensing", *9th International PiezoMEMS Workshop*, 2021.
4. A. Ravi, **V. Gund**, A. Lal, "Acousto-Optic Modulation in TiO_2 crystal Using GHz Fresnel-type Focusing Ultrasonic Transducer", *IEEE International Ultrasonics Symposium (IUS)*, 2021.

5. J. Casamento, **V. Gund**, K. Nomoto, T. Maeda, B. Davaji, M. J. Asadi, J. Wright, Y. Shao, D. A. Muller, A. Lal, H. Xing, and D. Jena, "Epitaxial $Sc_xAl_{1-x}N$: Structural, Chemical, Electrical Properties, and Ferroelectric Behavior", *21st International Conference on Molecular Beam Epitaxy*, 2021.
6. H. Lee, B. Davaji, T. Maeda, **V. Gund**, J. Casamento, K. Nomoto, X. Li, Z. Zhang, A. Lal, D. Jena, and H. G. Xing, "Design Considerations For ScAlN/AlGaIn/GaN Ferroelectric HEMT", *Compound Semiconductor Week*, 2021.

PATENTS

1. **V. Gund**, "Apparatus and Method for Adjusting Elasticity of a Spring-Mass System," (*US012687394B2*)
 2. **V. V. Gund**, K. P. O'Brien, K. Jun, E. Mohammed, A. Sen Gupta, M. V. Metz, I. L. Ban, P. Fischer, "Digitally controlled lithographically-defined multi-frequency acoustic resonators," (*US012375060B2*)
 3. A. Lal, S. Jadhav, **V. Gund**, B. Davaji, G. Xing, D. Jena, "Resistive electrodes on ferroelectric devices for linear piezoelectric programming," (*US012051474B2*)
 4. A. Lal and **V. Gund**, "Integrated Circuits Based Biosensors," (*US Patent 9683967*)
 5. S. Tin, J. Kriz, J. Ridley, A. Lal, C. K. Ober, S. Ardanuc, **V. Gund**, A. Ruyack, K. Camera, "Self-Destructing Chip" (*US Patent 9842812*)
- Filed Invention Disclosures
6. **V. Gund**, "MEMS Device With Distributed Programmable Elements," (*US 20260176130A1*)
 7. A. R. Brown, **V. Gund**, J. Reinke, M. H. Gnerlich, "Lorentz Force Velocimeter," (*US 20260160555A1*)
 8. **V. Gund**, "Phase Change Material Microelectromechanical Systems Based Analog Memory and Computational Device," (*US 20250048657A1*)
 9. J. Casamento, **V. Gund**, D. Jena, H.L. Lee, B. Davaji, A. Lal, H. Xing, T. Maeda, "Epitaxial $Sc_xAl_{1-x}N$ Semiconductor Devices," (*US 20230378276A1*)
 10. H. Alavi, I. Ban, T. Kamgaing, E. Mohammed, H. W. Then, K. Obrien, P. Fischer, J. E. Pelaez, **V. Gund**, "Broadband Acoustic Wave Resonator (AWR) Filters," (*US 20210194459A1*)
 11. **V. Gund**, M. H. Gnerlich, P. Cousseau, "Generating a Random Bitstream Output for an Inertial Measurement Unit," *Patent Filed 07/18/2025*
 12. **V. Gund**, M. H. Gnerlich, P. Cousseau, "Generating a Unique Unclonable Identifier from an Inertial Measurement Unit," *Patent Filed 06/19/2025*
 13. A. R. Brown, M. H. Gnerlich, **V. Gund**, J. Reinke "Using a Test Tone to Calibrate a Scale Factor for a Microelectromechanical System (MEMS) Gyroscope" *Patent Filed 04/16/2025*
 14. A. R. Brown, M. H. Gnerlich, **V. Gund**, J. Reinke "Techniques for Reducing Variations in a Difference Between Drive Mode and Sense Mode Frequency in a Split Mode Gyroscope" *Patent Filed 01/31/2025*

INVITED TALKS

1. "Microelectromechanical Systems (MEMS) Sensor Design", *Birla Institute of Technology and Science (BITS) Pilani, Goa Campus*, September 22, 2025.
2. "Microelectromechanical Systems (MEMS) for Environmentally Sustainable Manufacturing, Functionality, and Design", *Wesleyan University*, March 21, 2025.
3. "Microelectromechanical Systems (MEMS) for Environmentally Sustainable Manufacturing, Functionality, and Design", *Swarthmore College*, February 10, 2025.

4. "Introduction to MEMS", *CNF Technology & Characterization at the Nanoscale (CNF TCN) Course*, June 10, 2022.
5. "Sustainable Manufacturing, Multi-Functionality, and System Design of Microelectromechanical Systems (MEMS) for Internet-of-Things (IoT) Sensor Nodes", *University of Maine*, May 2, 2022.
6. "Microelectromechanical Systems (MEMS) for Environmentally Sustainable Manufacturing, Functionality, and Design", *Oregon State University*, March 31, 2022.
7. "Microelectromechanical Systems (MEMS) for Environmentally Sustainable Manufacturing, Functionality, and Design", *Western Washington University*, January 28, 2022.
8. "Transient MEMS for Vanishing Programmable Resources", *Cornell Nanoscale Facility (CNF) Annual Meeting*, September 15, 2015.

SOFTWARE AND EDA

Programming	C, C++, Java, MATLAB, Python
Layout	L-Edit, K-Layout
FEA Tools	ANSYS Mechanical, COMSOL, OnScale
Circuit Design	Cadence/Synopsys Tools, LTSpice, Eagle
Prototyping	ANSYS Discovery, Fusion360, MEMSPRO, Blender
Machine Learning	MATLAB implementation
Typesetting Document	LaTeX

DEVICE FABRICATION AND TESTING

Thin-film Deposition	Chemical Vapor Deposition, Sputtering, Evaporation, Electroplating
Lithography	Spin Coating, Solution Casting, Optical, Electron Beam
Etching	Reactive Ion Etching, Vapor HF, XeF_2 Si Etch, Ion Mill, Wet Bench
Microscopy	Optical, SEM, STM, AFM
Electrical Testing	Keithley 2400 and 4200 DC IV, RF Variable Network Analyzer
Mechanical Test	Laser Doppler Vibrometry, Shaker Table, Rate Table & Thermal Chamber
Other Proficiencies	Rapid Thermal Anneal, Dicing Saw, Profilometry, Wirebonder

SELECTED PRESS

1. Popular Science, *These self-destructing electronics can turn your data to dust on command*, 2018
2. Communications of the ACM, *Transient Electronics Take Shape*, 2018
3. Tech Explorist, *Scientists created new architecture for vaporizable electronics*, 2018
4. The Engineer, *Technique triggers electronic circuits to self-destruct*, 2018
5. Gov Tech, *What causes these circuits to self-destruct?*, 2018
6. New Atlas, *Circuits self-destruct in response to radio waves*, 2018
7. Cornell Chronicle, *Engineers create new architecture for vaporizable electronics*, 2018
8. Cornell Sun, *Cornell Researchers Create Tech Designed to Disappear into Thin Air*, 2017